

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JnanaSangama”, Belgaum -590014, Karnataka.



LAB REPORT

On

OPERATING SYSTEMS (23CS4PCOPS)

Submitted by

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**in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING**

**in
COMPUTER SCIENCE AND ENGINEERING**



**B.M.S. COLLEGE OF ENGINEERING
(Autonomous Institution under VTU)**

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CERTIFICATE

**B. M. S. College of Engineering,
Bull Temple Road, Bangalore 560019
(Affiliated To Visvesvaraya Technological University, Belgaum)
Department of Computer Science and Engineering**



This is to certify that the Lab work entitled “**OPERATING SYSTEMS**” carried out by ANANYA RANJITH (**1BM24CS041**), who is a bonafide student of **B. M. S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2025-26. The Lab report has been approved as it satisfies the academic requirements in respect of Operating Systems Lab - (**23CS4PCOPS**) work prescribed for the said degree.

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Course outcomes:

CO1	Analyze time complexity of algorithms using asymptotic notations.
CO2	Apply various algorithm design techniques for the given problem.
CO3	Evaluate the appropriate algorithm/design technique for solving a given problem.
CO4	Conduct practical experiments by applying appropriate algorithm design techniques to solve problems.

Lab program 1:

Write a C program to simulate the following CPU scheduling algorithm to find turnaround time and waiting time.

a) FCFS b) SJF c) Priority d) Round Robin

a) FCFS

```
#include <stdio.h>

int main() {
    int n, i;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    int bt[n], wt[n], tat[n];
    printf("Enter Burst Times:\n");
    for(i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }
    wt[0] = 0;
    for(i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }
    for(i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }
    float avg_wt = 0, avg_tat = 0;
    printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");
    for(i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
        avg_wt += wt[i];
        avg_tat += tat[i];
    }
    avg_wt /= n;
    avg_tat /= n;
    printf("\nAverage Waiting Time = %.2f", avg_wt);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
    return 0;
}
```

Output:

```

Default: C:\Users\anany\Desktop\OS\FCFS.exe
ctrl+alt+t

C:\Users\anany\Desktop\OS\ >
Enter number of processes: 4
Enter Burst Times:
P1: 5
P2: 3
P3: 8
P4: 6

Process Burst Time    Waiting Time    Turnaround Time
P1      5             0              5
P2      3             5              8
P3      8             8              16
P4      6             16              22

Average Waiting Time = 7.25
Average Turnaround Time = 12.75

Process returned 0 (0x0)   execution time : 17.902 s
Press any key to continue.

```

Observation:

OS lab

FCFS

```

#include <stdio.h>
struct process {
    int pid;
    int at;
    int bt;
    int ct;
    int tat;
    int wt;
};

int main() {
    int n, i;
    int c_time = 0;
    int total_wt = 0, total_tat = 0;
    printf("Enter no. of processes: ");
    scanf("%d", &n);
    struct process p[10];
    for(i=0; i<n; i++) {
        printf("Enter AT and BT for process %d: ", i+1);
        scanf("%d %d", &p[i].at, &p[i].bt);
    }
    for(i=0; i<n; i++) {
        for(j=i+1; j<n; j++) {
            if(p[j].at < p[i].at) {
                temp = p[i];
                p[i] = p[j];
                p[j] = temp;
            }
        }
    }

```

SURYA GURU

```

for(i=0; i<n; i++) {
    if(c_time < p[i].at)
        c_time = p[i].at;
    p[i].wt = c_time + p[i].bt;
    p[i].tat = p[i].wt + p[i].at;
    p[i].wt = p[i].wt - p[i].bt;
    c_time = p[i].wt;
    total_wt += p[i].wt;
    total_tat += p[i].tat;
}

printf("\n PID \t AT \t BT \t CT \t TAT \t WT \n");
for(i=0; i<n; i++) {
    printf("%d \t %d \t %d \t %d \t %d \t %d \n",
        p[i].pid,
        p[i].at,
        p[i].bt,
        p[i].ct,
        p[i].tat,
        p[i].wt);
}

printf("\n Average WT = %f", (float) total_wt / n);
printf("\n Average TAT = %f", (float) total_tat / n);
return 0;

```

PID	AT	BT	CT	TAT	WT
1	6	8	14	8	0
2	6	8	22	16	8
3	7	9	31	24	15

Average WT = 7.67

Average TAT = 16.00

OUTPUT

PID	AT	BT	CT	TAT	WT
1	0	5	5	5	5
2	0	3	8	8	5
3	0	8	16	16	8

Average WT = 9.67

Average TAT = 9.67

b) SJF

```
#include <stdio.h>

int main() {
    int n, i, j;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    int bt[n], wt[n], tat[n], p[n];
    printf("Enter Burst Times:\n");
    for(i = 0; i < n; i++) {
        p[i] = i + 1;
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }
    for(i = 0; i < n - 1; i++) {
        for(j = i + 1; j < n; j++) {
            if(bt[i] > bt[j]) {
                int temp;
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
                temp = p[i];
                p[i] = p[j];
                p[j] = temp;
            }
        }
    }
    wt[0] = 0;
    for(i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }
    for(i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }
    float avg_wt = 0, avg_tat = 0;
    printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");
    for(i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            p[i], bt[i], wt[i], tat[i]);
        avg_wt += wt[i];
        avg_tat += tat[i];
    }
}
```

```

avg_wt /= n;
avg_tat /= n;
printf("\nAverage Waiting Time = %.2f", avg_wt);
printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
return 0;
}

```

Output:

```

Default: C:\Users\anany\Desktop\OS\SJF.exe
ctrl+alt+j

C:\Users\anany\Desktop\OS\SJF.exe

Enter number of processes: 4
Enter Burst Times:
P1: 6
P2: 8
P3: 7
P4: 3

Process Burst Time    Waiting Time    Turnaround Time
P4      3             0              3
P1      6             3              9
P3      7             9             16
P2      8            16             24

Average Waiting Time = 7.00
Average Turnaround Time = 13.00

Process returned 0 (0x0)   execution time : 18.303 s
Press any key to continue.

```

Observation:

Left Page (Code):

```

#include <stdio.h>
#define process 4
int pid,
int bt,
int at,
int ct,
int tat,
int wt,
int min_bt,
int remaining_bt,
int completed;

void NonPreemptive (struct Process p[4], int n) {
    int current_time = 0, completed = 0;
    float total_wt = 0, total_tat = 0;
    while (completed < n) {
        int index = -1;
        int min_bt = 9999;
        for (int i = 0; i < n; i++) {
            if (p[i].ct < current_time && p[i].completed == 0) {
                if (p[i].bt < min_bt) {
                    min_bt = p[i].bt;
                    index = i;
                }
            }
        }
        if (index != -1) {
            current_time += p[index].bt;
            p[index].ct = current_time;
            p[index].tat = p[index].ct - p[index].at;
            p[index].wt = p[index].tat - p[index].bt;
            p[index].completed = 1;
            completed++;
        }
    }
}

```

Right Page (Output and Gantt Chart):

OUTPUT:

```

Enter number of processes: 4
Process P1
Arrival Time: 0
Burst Time: 6
Process P2
Arrival Time: 0
Burst Time: 8
Process P3
Arrival Time: 0
Burst Time: 7
Process P4
Arrival Time: 5
Burst Time: 3

```

Gantt Chart:

Process	Arrival Time	Burst Time	Completion Time	Waiting Time	Turnaround Time
P1	0	6	6	3	9
P3	0	7	13	9	16
P4	5	3	16	11	16
P2	0	8	24	16	24

Average Waiting Time: 7.00 ms
Average TAT: 13.00 ms

c) Priority

```
#include <stdio.h>

int main() {
    int n, i, j;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    int bt[n], pr[n], wt[n], tat[n], p[n];
    printf("Enter Burst Time and Priority:\n");
    for(i = 0; i < n; i++) {
        p[i] = i + 1;
        printf("P%d Burst Time: ", i + 1);
        scanf("%d", &bt[i]);
        printf("P%d Priority: ", i + 1);
        scanf("%d", &pr[i]);
    }
    for(i = 0; i < n - 1; i++) {
        for(j = i + 1; j < n; j++) {
            if(pr[i] > pr[j]) {
                int temp;
                temp = pr[i];
                pr[i] = pr[j];
                pr[j] = temp;
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
                temp = p[i];
                p[i] = p[j];
                p[j] = temp;
            }
        }
    }
    wt[0] = 0;
    for(i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }
    for(i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }
    float avg_wt = 0, avg_tat = 0;
    printf("\nProcess\tPriority\tBurst Time\tWaiting Time\tTurnaround Time\n");
```



```

for(i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i], pr[i], bt[i], wt[i], tat[i]);
    avg_wt += wt[i];
    avg_tat += tat[i];
}
avg_wt /= n;
avg_tat /= n;
printf("\nAverage Waiting Time = %.2f", avg_wt);
printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
return 0;
}

```

Output:

```

Default: "C:\Users\anany\Desktop\OS\Priority
scheduling.exe"
ctrl+alt+1

Enter number of processes: 4
Enter Burst Time and Priority:
P1 Burst Time: 10
P1 Priority: 3
P2 Burst Time: 5
P2 Priority: 1
P3 Burst Time: 8
P3 Priority: 4
P4 Burst Time: 6
P4 Priority: 2

Process Priority      Burst Time      Waiting Time      Turnaround Time
P2         1             5                0                 5
P4         2             6                5                11
P1         3            10               11               21
P3         4             8               21               29

Average Waiting Time = 9.25
Average Turnaround Time = 16.50

Process returned 0 (0x0)   execution time : 21.663 s
Press any key to continue.
|

```

10 | Page

Handwritten code for Priority Scheduling in C++.

```
#include <iostream>
using namespace std;

int p[10];
int at[10];
int bt[10];
int pr[10];
int ct[10];
int tot[10];
int kty;
int kty;
int done[10];

void calculate( struct Process p[], int n ) {
    float total = 0, total_bt = 0;
    for (int i = 0; i < n; i++) {
        p[i].bt = p[i].at - p[i].pr;
        total += p[i].bt;
        total_bt += p[i].bt;
    }
    cout << "Average AT = " << total / n << endl;
    cout << "Average BT = " << total_bt / n << endl;
}

void nonPreemptive( struct Process p[], int n ) {
    int completed = 0, time = 0;
    while (1) {
        for (int i = 0; i < n; i++) {
            if (p[i].at <= time && p[i].done == 0) {
                p[i].pr = p[i].bt;
                p[i].done = 1;
                completed++;
            }
        }
        if (completed == n) break;
        time++;
    }
}
```

Output:

```
Enter number of Processes : 3
Enter choices :
Process P1: 10, 20, 30, 10, 10, 0
Process P2: 5, 5, 5, 5, 5, 5
Process P3: 1, 1, 1, 1, 1, 1
Average AT: 6.33 ms
Average BT: 10 ms
Enter choices:
```

d) Round Robin

```
#include <stdio.h>

int main() {
    int n, i, quantum;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    int bt[n], rem_bt[n], wt[n], tat[n];
    printf("Enter Burst Times:\n");
    for(i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
        rem_bt[i] = bt[i];
    }
    printf("Enter Time Quantum: ");
    scanf("%d", &quantum);
    int time = 0;
    while(1) {
        int done = 1;
        for(i = 0; i < n; i++) {
            if(rem_bt[i] > 0) {
                done = 0;
                if(rem_bt[i] > quantum) {
                    time += quantum;
                    rem_bt[i] -= quantum;
                }
                else {
                    time += rem_bt[i];
                    wt[i] = time - bt[i];
                    rem_bt[i] = 0;
                }
            }
        }
        if(done)
            break;
    }
    float avg_wt = 0, avg_tat = 0;
    printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");
    for(i = 0; i < n; i++) {
        tat[i] = bt[i] + wt[i];
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }
}
```

```

        avg_wt += wt[i];
        avg_tat += tat[i];
    }
    avg_wt /= n;
    avg_tat /= n;
    printf("\nAverage Waiting Time = %.2f", avg_wt);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
    return 0;
}

```

Output:

```

C:\Users\anany\Desktop\OS\RR.exe
ctrl+alt+1

Enter number of processes: 4
Enter Burst Times:
P1: 10
P2: 5
P3: 8
P4: 6
Enter Time Quantum: 6

Process Burst Time      Waiting Time      Turnaround Time
P1      10              17                27
P2       5               6                 11
P3       8              21                 29
P4       6              17                 23

Average Waiting Time = 15.25
Average Turnaround Time = 22.50

Process returned 0 (0x0)   execution time : 19.402 s
Press any key to continue.
|

```


Observation:

Round Robin

```

#include <stdio.h>
struct Process {
    int pid;
    int at;
    int bt;
    int st;
    int et;
    int tat;
    int wt;
};

void calculate(struct Process p[], int n) {
    float total_wt = 0, total_tat = 0;
    printf("PID | AT | BT | CT | TAT | WT | \n");
    for (int i = 0; i < n; i++) {
        p[i].wt = p[i].at - p[i].st;
        p[i].tat = p[i].tat + p[i].bt;
        total_wt += p[i].wt;
        total_tat += p[i].tat;
    }
    printf("P | AT | BT | CT | TAT | WT | \n");
    printf("P1 | 0 | 5 | 13 | 13 | 8 | \n");
    printf("P2 | 1 | 3 | 12 | 11 | 8 | \n");
    printf("P3 | 2 | 1 | 5 | 3 | 2 | \n");
    printf("P4 | 3 | 2 | 9 | 6 | 4 | \n");
    printf("P5 | 4 | 3 | 14 | 10 | 7 | \n");
    printf("Average WT = %.2f, total_wt/n);", total_wt/n);
    printf("Average TAT = %.2f, total_tat/n);", total_tat/n);
}

void RoundRobin(struct Process p[], int n, int tq) {
    int time = 0, completed = 0;
    int queue[100], front = 0, rear = 0;
    int visited[100] = {0};
    for (int i = 0; i < n; i++) {
        if (p[i].at == 0) {
            queue[rear++] = i;
        }
    }

    int main() {
        int n, tq;
        printf("Enter number of processes: ");
        scanf("%d", &n);
        struct Process p[n];
        for (int i = 0; i < n; i++) {
            p[i].pid = i+1;
            printf("Process %d, ", i+1);
            printf("Arrival time: ");
            scanf("%d", &p[i].at);
            printf("Burst Time: ");
            scanf("%d", &p[i].bt);
            p[i].st = p[i].bt;
        }
        printf("\nEnter Time Quantum: ");
        scanf("%d", &tq);
        RoundRobin(p, n, tq);
        calculate(p, n);
        return 0;
    }
    
```

Output:

Enter number of processes: 5

Process P1
Arrival time = 0
Burst time = 5

Process P2
Arrival time = 1
Burst time = 3

Process P3
Arrival time = 2

```

visited[i] = 1;
}

while (completed < n) {
    if (front == rear) {
        time++;
        for (int i = 0; i < n; i++) {
            if (p[i].at <= time && !visited[i]) {
                queue[rear++] = i;
                visited[i] = 1;
            }
        }
        continue;
    }
    int i = queue[front++];
    if (p[i].at > time) {
        time = p[i].at;
        p[i].st = time;
        completed++;
    } else {
        time += p[i].bt;
        p[i].st = 0;
        p[i].ct = time;
        completed++;
    }
    for (int j = 0; j < n; j++) {
        if (p[j].at <= time && !visited[j]) {
            queue[rear++] = j;
            visited[j] = 1;
        }
    }
    if (p[i].bt > 0) {
        Burst time = 1
        Process P4
        Arrival time = 3
        Burst time = 2

        Process P5
        Arrival time = 4
        Burst time = 3
        Enter time quantum = 2

        Process AT BT CT TAT WT
        P1 0 5 13 13 8
        P2 1 3 12 11 8
        P3 2 1 5 3 2
        P4 3 2 9 6 4
        P5 4 3 14 10 7

        P1 P2 P3 P4 P5 P2 P1 P5
        0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

        Average WT = 5.8 ms
        Average TAT = 8.6 ms
    }
}
    
```

Lab program 2:

Write a C program to simulate a multi-level queue scheduling algorithm considering the following scenario. All the processes in the system are divided into two categories – system processes and user processes. System processes are to be given higher priority than user processes. Use FCFS scheduling for the processes in each queue.

```
#include <stdio.h>
struct Process {
    int id;
    int burst_time;
    int waiting_time;
    int turnaround_time;
};
void fcfs(struct Process p[], int n) {
    int i;
    p[0].waiting_time = 0;
    for (i = 1; i < n; i++) {
        p[i].waiting_time = p[i - 1].waiting_time + p[i - 1].burst_time;
    }
    for (i = 0; i < n; i++) {
        p[i].turnaround_time = p[i].waiting_time + p[i].burst_time;
    }
}
void display(struct Process p[], int n, char name[]) {
    int i;
    float total_wt = 0, total_tat = 0;
    printf("\n%s Queue:\n", name);
    printf("Process\tBurst Time\tWaiting Time\tTurnaround Time\n");
    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            p[i].id,
            p[i].burst_time,
            p[i].waiting_time,
            p[i].turnaround_time);
        total_wt += p[i].waiting_time;
        total_tat += p[i].turnaround_time;
    }
    printf("Average Waiting Time = %.2f\n", total_wt / n);
    printf("Average Turnaround Time = %.2f\n", total_tat / n);
}
int main() {
```

```

int n_sys, n_user, i;
printf("Enter number of System Processes: ");
scanf("%d", &n_sys);
struct Process sys[n_sys];
for (i = 0; i < n_sys; i++) {
    sys[i].id = i + 1;
    printf("Enter burst time for System Process P%d: ", sys[i].id);
    scanf("%d", &sys[i].burst_time);
}
printf("\nEnter number of User Processes: ");
scanf("%d", &n_user);
struct Process user[n_user];
for (i = 0; i < n_user; i++) {
    user[i].id = i + 1;
    printf("Enter burst time for User Process P%d: ", user[i].id);
    scanf("%d", &user[i].burst_time);
}
fcfs(sys, n_sys);
fcfs(user, n_user);
int total_sys_time = 0;
for (i = 0; i < n_sys; i++) {
    total_sys_time += sys[i].burst_time;
}
for (i = 0; i < n_user; i++) {
    user[i].waiting_time += total_sys_time;
    user[i].turnaround_time = user[i].waiting_time + user[i].burst_time;
}
printf("\n===== Multi-Level Queue Scheduling =====\n");
display(sys, n_sys, "System (Higher Priority)");
display(user, n_user, "User (Lower Priority)");
return 0;
}

```

Output:

```
Default: "C:\Users\RMSCECSE-14\Desktop\nM24-41\MultiLevel Queueing.exe"  
ctrl+alt+f  
Open with ▾  
"C:\Users\RMSCECSE-14\Desktop\nM24-41\MultiLevel Queueing.exe" X + ▾  
Enter number of System Processes: 4  
Enter burst time for System Process P1: 4  
Enter burst time for System Process P2: 5  
Enter burst time for System Process P3: 3  
Enter burst time for System Process P4: 6  
  
Enter number of User Processes: 4  
Enter burst time for User Process P1: 8  
Enter burst time for User Process P2: 9  
Enter burst time for User Process P3: 5  
Enter burst time for User Process P4: 7  
  
===== Multi-Level Queue Scheduling =====  
  
System (Higher Priority) Queue:  
Process Burst Time Waiting Time Turnaround Time  
P1 4 0 4  
P2 5 4 9  
P3 3 9 12  
P4 6 12 18  
Average Waiting Time = 6.25  
Average Turnaround Time = 10.75  
  
User (Lower Priority) Queue:  
Process Burst Time Waiting Time Turnaround Time  
P1 8 18 26  
P2 9 26 35  
P3 5 35 40  
P4 7 40 47  
Average Waiting Time = 29.75  
Average Turnaround Time = 37.00  
  
Process returned 0 (0x0) execution time : 194.541 s  
Press any key to continue.  
|
```


Observation:

```

Multilevel queuing
#include <stdio.h>
struct Process {
    int id;
    int burst-time;
    int waiting-time;
    int tat;
};

void fcts (struct Process p[], int n) {
    int i;
    p[0].waiting-time = 0;
    for (i = 1; i < n; i++) {
        p[i].waiting-time = p[i-1].waiting-time +
                           p[i-1].burst-time;
    }
    for (i = 0; i < n; i++) {
        p[i].tat = p[i].waiting-time + p[i].burst-time;
    }
}

void display (struct Process p[], int n, char name[]) {
    int i;
    float total-wt = 0, total-tat = 0;
    printf("\n %s Queue: %s\n", name);
    printf("Process | BT | WT | TAT | \n");
    for (i = 0; i < n; i++) {
        printf("P %d | %d | %d | %d | \n",
               p[i].id, p[i].burst-time,
               p[i].waiting-time, p[i].tat);
        total-wt += p[i].waiting-time;
    }
}

```

Output:

Process	Queue	AT	BT
P1	↓	0	4
P2	System Queue	1	5
P3		2	3
P4		3	6

Process	Queue	AT	BT
P1	↓	0	8
P2	User Queue	1	9
P3		2	5
P4		3	7

Enter number of System Processes: 4
 Enter BT for P1: 4
 Enter BT for P2: 5
 Enter BT for P3: 3
 Enter BT for P4: 6

Enter number of User Processes: 4
 Enter BT for P1: 8
 Enter BT for P2: 9
 Enter BT for P3: 5
 Enter BT for P4: 7

Multilevel queue Scheduling

System (Higher Priority) Queue:

```

int main() {
    int n-sys, n-user, i;
    printf("Enter no of System Processes: ");
    scanf("%d", &n-sys);
    struct Process user[n-user];
    for (i = 0; i < n-user; i++) {
        user[i].id = i+1;
        printf("Enter burst-time for User Process\n Process P%d:", user[i].id);
        scanf("%d", &user[i].burst-time);
    }
    fcts(sys, n-sys);
    fcts(user, n-user);
    int total-sys-time = 0;
    for (i = 0; i < n-sys; i++) {
        total-sys-time += sys[i].burst-time;
    }
    for (i = 0; i < n-user; i++) {
        user[i].waiting-time += total-sys-time;
        user[i].tat = user[i].waiting-time +
                     user[i].burst-time;
    }
    printf("Multilevel Queue Scheduling");
    display(sys, n-sys, "System (Higher Priority)");
    display(user, n-user, "User (Lower Priority)");
}

```

Process	BT	WT	TAT
P1	4	0	4
P2	5	4	9
P3	3	9	12
P4	6	12	18

Average Waiting Time = 6.25 ms
 Average Turnaround Time = 10.75 ms

User (Lower Priority) Queue:

Process	BT	WT	TAT
P1	8	18	26
P2	9	26	35
P3	5	35	40
P4	7	40	47

Average Waiting Time = 29.75 ms
 Average Turnaround Time = 37 ms

P1	P2	P3	P4	P1	P2	P3	P4
0	4	9	12	18	26	35	46

$\frac{P_1}{10/125}$

Lab program 3:

Write a C program to simulate Real-Time CPU Scheduling algorithms:

a) Rate- Monotonic b) Earliest-deadline First c) Proportional scheduling

a) Rate- Monotonic

```
#include <stdio.h>
int main() {
    int n, i, j;
    printf("Enter number of tasks: ");
    scanf("%d", &n);
    int task[n], period[n], burst[n];
    for(i = 0; i < n; i++) {
        task[i] = i + 1;
        printf("\nTask %d Execution Time: ", i + 1);
        scanf("%d", &burst[i]);
        printf("Task %d Period: ", i + 1);
        scanf("%d", &period[i]);
    }
    for(i = 0; i < n - 1; i++) {
        for(j = i + 1; j < n; j++) {
            if(period[i] > period[j]) {
                int temp;
                temp = period[i];
                period[i] = period[j];
                period[j] = temp;
                temp = burst[i];
                burst[i] = burst[j];
                burst[j] = temp;
                temp = task[i];
                task[i] = task[j];
                task[j] = temp;
            }
        }
    }
    printf("\n\nRate Monotonic Scheduling (RMS)\n");
    printf("Task\tExecution Time\tPeriod\tPriority\n");
    for(i = 0; i < n; i++) {
        printf("T%d\t%d\t%d\t%d\n",
            task[i], burst[i], period[i], i + 1);
    }
}
```

```

}
printf("\nExecution Sequence:\n");
for(i = 0; i < n; i++) {
    printf("T%d", task[i]);
    if(i != n - 1)
        printf(" -> ");
}
printf("\n");
return 0;
}

```

Output:

```

C:\Users\anany\Desktop\OS\RMS.exe
ctrl+alt+1

Enter number of tasks: 4

Task 1 Execution Time: 1
Task 1 Period: 8

Task 2 Execution Time: 2
Task 2 Period: 4

Task 3 Execution Time: 1
Task 3 Period: 12

Task 4 Execution Time: 2
Task 4 Period: 6

Rate Monotonic Scheduling (RMS)
Task    Execution Time  Period  Priority
T2      2              4       1
T4      2              6       2
T1      1              8       3
T3      1              12      4

Execution Sequence:
T2 -> T4 -> T1 -> T3

Process returned 0 (0x0)   execution time : 24.548 s
Press any key to continue.

```

Observation:

Rate Monotonic Scheduling

```
#include <stdio.h>
struct Process {
    int id, period, burst, remaining, next arrival;
};

int main() {
    int n, time, type = 0;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    struct Process p[n];
    for (int i = 0; i < n; i++) {
        printf("Process %d (Burst Time, Period): ", i+1);
        scanf("%d %d", &p[i].burst, &p[i].period);
        p[i].remaining = p[i].burst;
        p[i].next arrival = 0;
        p[i].id = i+1;
        if (p[i].period < type)
            type = p[i].period;
    }
    printf("\n PMS\n");
    printf("Time to Running\n");
    for (int i = 0; i < n; i++)
        printf("%d\n", p[i].id);
    for (time = 0; time < type * 2; time++) {
        int highest = -1;
        for (int i = 0; i < n; i++) {
            if (time == p[i].next arrival) {
                p[i].remaining = p[i].burst;
                p[i].next arrival = p[i].period;
            }
            if (p[i].remaining > 0) {
                if (highest == -1 || p[i].period < p[highest].period)
                    highest = i;
            }
        }
        printf("%d\n", p[highest].id);
        p[highest].remaining--;
        if (p[highest].remaining == 0)
            printf("Idle\n");
    }
    return 0;
}
```

OUTPUT:

Enter number of processes 4
 Process 1 (Burst Time, Period): 6 8
 Process 2 (Burst Time, Period): 4 2
 Process 3 (Burst Time, Period): 2 4
 Process 4 (Burst Time, Period): 2 3

PMS

Time	Running	P1.rem	P2.rem	P3.rem	P4.rem
0	P1	6	4	2	1
1	P1	6	4	2	0
2	P3	6	4	1	0
3	P4	6	4	1	1
4	P4	6	4	1	0
5	P3	6	4	1	0

Gantt chart:

P1	P2	P3	P4	P3	P4	P3	P4	P3	P4
0	1								

b) Earliest-deadline First

```
#include <stdio.h>

int main() {
    int n, i, j;
    printf("Enter number of tasks: ");
    scanf("%d", &n);
    int task[n], burst[n], deadline[n];
    for(i = 0; i < n; i++) {
        task[i] = i + 1;
        printf("\nTask %d Execution Time: ", i + 1);
        scanf("%d", &burst[i]);
        printf("Task %d Deadline: ", i + 1);
        scanf("%d", &deadline[i]);
    }
    for(i = 0; i < n - 1; i++) {
        for(j = i + 1; j < n; j++) {
            if(deadline[i] > deadline[j]) {
                int temp;
                temp = deadline[i];
                deadline[i] = deadline[j];
                deadline[j] = temp;
                temp = burst[i];
                burst[i] = burst[j];
                burst[j] = temp;
                temp = task[i];
                task[i] = task[j];
                task[j] = temp;
            }
        }
    }
    printf("\n\nEarliest Deadline First (EDF) Scheduling\n");
    printf("Task\tExecution Time\tDeadline\tPriority\n");
    for(i = 0; i < n; i++) {
        printf("T%d\t%d\t%d\t%d\n",
            task[i], burst[i], deadline[i], i + 1);
    }
    int currentTime = 0;
    printf("\nDeadline Status:\n");
    printf("Task\tCompletion Time\tStatus\n");
```



```

for(i = 0; i < n; i++) {
    currentTime += burst[i];
    if(currentTime <= deadline[i])
        printf("T%d\t%d\t\tMet\n", task[i], currentTime);
    else
        printf("T%d\t%d\t\tMissed\n", task[i], currentTime);
}
printf("\nExecution Sequence:\n");
for(i = 0; i < n; i++) {
    printf("T%d", task[i]);
    if(i != n - 1)
        printf(" -> ");
}
printf("\n");
return 0;
}

```

Output:

```

C:\Users\anany\Desktop\OS\EDF.exe
Enter number of tasks: 4
Task 1 Execution Time: 2
Task 1 Deadline: 8
Task 2 Execution Time: 1
Task 2 Deadline: 4
Task 3 Execution Time: 3
Task 3 Deadline: 12
Task 4 Execution Time: 2
Task 4 Deadline: 6

Earliest Deadline First (EDF) Scheduling
Task    Execution Time    Deadline    Priority
T2      1                4           1
T4      2                6           2
T1      2                8           3
T3      3                12          4

Deadline Status:
Task    Completion Time    Status
T2      1                Met
T4      3                Met
T1      5                Met
T3      8                Met

Execution Sequence:
T2 -> T4 -> T1 -> T3

Process returned 0 (0x0)   execution time : 20.986 s
Press any key to continue.

```

Observation:

Earliest Deadline First

```
#include <stdio.h>
struct Task {
    int id;
    int capacity;
    int period;
    int deadline;
    int remaining;
};

int main() {
    int n, total_time;
    printf("Enter number of tasks:");
    scanf("%d", &n);
    Task tasks[n];

    for (int i = 0; i < n; i++) {
        tasks[i].id = i+1;
        printf("Enter Capacity and Period for Task %d:", i+1);
        scanf("%d %d", &tasks[i].capacity, &tasks[i].period);
        tasks[i].deadline = tasks[i].period;
        tasks[i].remaining = tasks[i].capacity;
    }

    printf("Enter total simulation time:");
    scanf("%d", &total_time);
    printf("\n Scheduling Log\n");
    for (int t = 0; t < total_time; t++) {
        int selected_task = -1;
        int min_deadline = 9999;

        for (int i = 0; i < n; i++) {
            if (tasks[i].remaining > 0) {
                if (tasks[i].deadline < min_deadline) {
                    min_deadline = tasks[i].deadline;
                    selected_task = i;
                }
            }
        }

        if (selected_task != -1) {
            printf("Time %d: Task %d is running\n", t, selected_task);
            tasks[selected_task].remaining--;
            tasks[selected_task].deadline--;
            if (tasks[selected_task].remaining == 0) {
                tasks[selected_task].deadline = tasks[selected_task].period;
                tasks[selected_task].remaining = tasks[selected_task].capacity;
            }
        } else {
            printf("Time %d: CPU Idle\n", t);
        }

        if (t % 10 == 9) {
            return 0;
        }
    }
}
```

OUTPUT

```
Enter number of processes: 3
Process 1: 1 4
Process 2: 2 6
Process 3: 3 8

PID    Burst time    Period
P1      1             4
P2      2             6
P3      3             8
```

Scheduling Log

Time	Scheduling
0	P1
1	P2
2	P3
3	P2
4	P1
5	Idle
6	P2
7	P3
8	P1
9	P3
10	Idle
11	P3

c) Proportional scheduling

```
#include <stdio.h>
int main() {
    int n, i;
    float totalWeight = 0, totalCPU;
    printf("Enter number of tasks: ");
    scanf("%d", &n);
    int task[n];
    float weight[n], cpuShare[n];
    printf("Enter Total CPU Time Available: ");
    scanf("%f", &totalCPU);
    for(i = 0; i < n; i++) {
        task[i] = i + 1;
        printf("\nEnter weight for Task %d: ", i + 1);
        scanf("%f", &weight[i]);
        totalWeight += weight[i];
    }
    for(i = 0; i < n; i++) {
        cpuShare[i] = (weight[i] / totalWeight) * totalCPU;
    }
    printf("\n\nProportional Scheduling\n");
    printf("Task\tWeight\t\tCPU Share\n");
    for(i = 0; i < n; i++) {
        printf("T%d\t%.2f\t\t%.2f\n",
            task[i], weight[i], cpuShare[i]);
    }
    printf("\nExecution Sequence:\n");
    for(i = 0; i < n; i++) {
        printf("T%d", task[i]);
        if(i != n - 1)
            printf(" -> ");
    }
    printf("\n");
    return 0;
}
```

Output:

```
Default: C:\Users\anany\Desktop\OS\prop-sch.exe
ctrl+alt+f

C:\Users\anany\Desktop\OS\>

Enter number of tasks: 3
Enter Total CPU Time Available: 100

Enter weight for Task 1: 50
Enter weight for Task 2: 30
Enter weight for Task 3: 20

Proportional Scheduling
Task   Weight   CPU Share
T1     50.00   50.00
T2     30.00   30.00
T3     20.00   20.00

Execution Sequence:
T1 -> T2 -> T3
```

Observation:

Proportional Sharing

```
#include <iostream>
int main() {
    int n, i;
    int burst[10], share[10];
    float cputime[10], totalBurst = 0;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    for (i = 0; i < n; i++) {
        printf("\n Process P%d \n", i+1);
        printf("Enter Burst Time: ");
        scanf("%d", &burst[i]);
        printf("Enter CPU Share (in %): ");
        scanf("%d", &share[i]);
        totalBurst += burst[i];
    }

    printf("\nProportional Sharing Scheduling \n");
    printf("Process | Burst Time | CPU Sharing | Allocated CPU Time \n");
    for (i = 0; i < n; i++) {
        cputime[i] = (share[i]/100.0) * totalBurst;
        printf("P%d | %d | %d% | %d \n", i+1, burst[i], share[i], cputime[i]);
    }

    return 0;
}
```

OUTPUT:

Enter number of processes: 4

Process P1

Enter Burst Time: 10

Enter CPU Share (in %): 20

Process P2

Enter Burst Time: 15

Enter CPU Share (in %): 30

Process P3

Enter Burst Time: 20

Enter CPU Share (in %): 25

Process P4

Enter Burst Time: 5

Enter CPU Share (in %): 25

Proportional Sharing Scheduling

Process	Burst Time	CPU Sharing	Allocated CPU Time
P1	10	20%	10.00
P2	15	30%	15.00
P3	20	25%	12.50
P4	5	25%	12.50

Total Burst Time = 10 + 15 + 20 + 5 = 50

$P_1 \rightarrow \frac{20}{100} \times 50 = 10$

$P_2 \rightarrow \frac{30}{100} \times 50 = 15$

$P_3 \rightarrow \frac{25}{100} \times 50 = 12.5$

$P_4 \rightarrow \frac{25}{100} \times 50 = 12.5$

Lab program 4:

Write a C program to simulate:

a) Producer-Consumer problem using semaphores. b) Dining-Philosopher's problem

a) Producer-Consumer problem using semaphores

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>
#define BUFFER_SIZE 8
int buffer[BUFFER_SIZE];
int in = 0;
int out = 0;
int totalItems;
sem_t empty;
sem_t full;
pthread_mutex_t mutex;
int produceItem(int item)
{
    return item;
}
void consumeItem(int item)
{
    printf("Consumed Item: %d\n", item);
}
void* producer(void* arg)
{
    int item;
    for (int i = 1; i <= totalItems; i++)
    {
        item = produceItem(i);
        sem_wait(&empty);
        pthread_mutex_lock(&mutex);
        buffer[in] = item;
        printf("Producer produced %d at position %d\n", item, in);
        in = (in + 1) % BUFFER_SIZE;
        pthread_mutex_unlock(&mutex);
        sem_post(&full);
        sleep(1);
    }
}
```

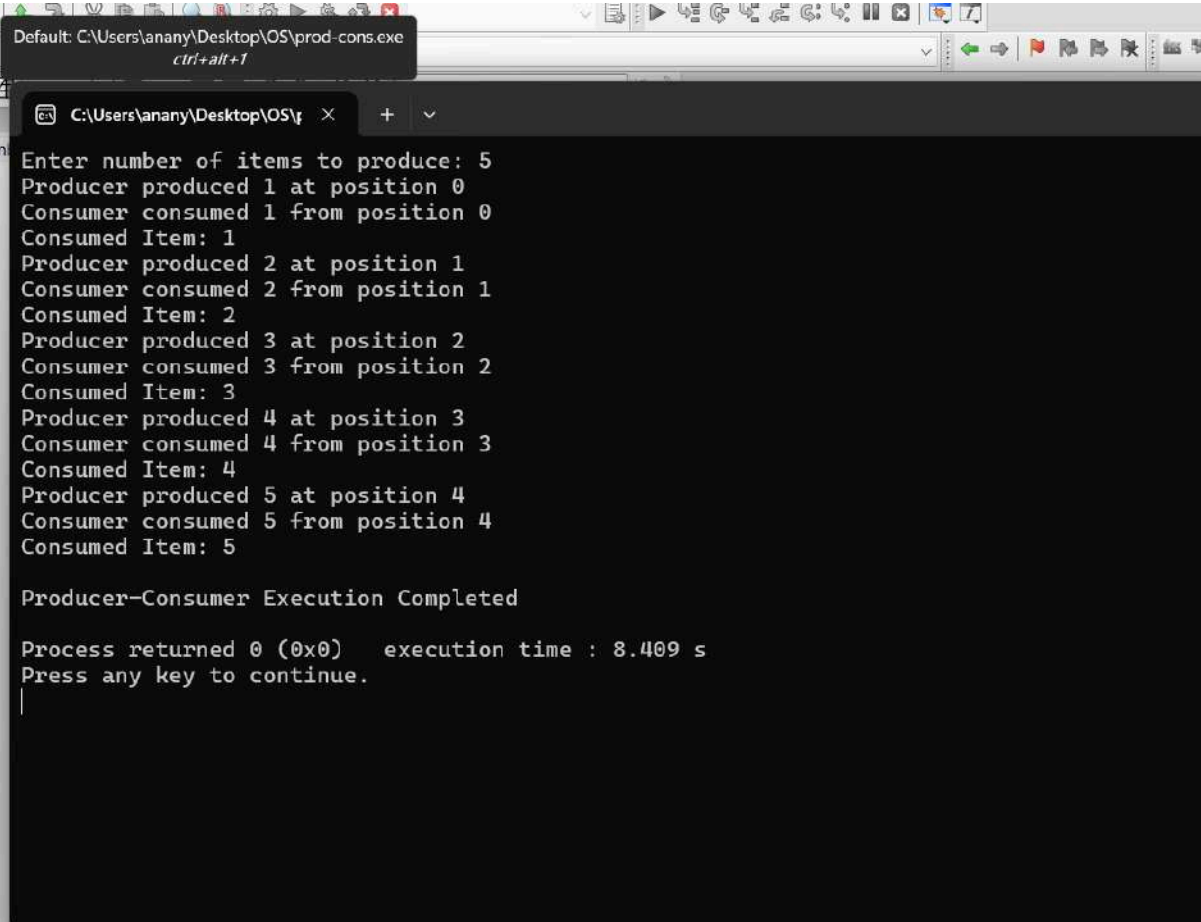
```

    pthread_exit(NULL);
}
void* consumer(void* arg)
{
    int item;
    for (int i = 1; i <= totalItems; i++)
    {
        sem_wait(&full);
        pthread_mutex_lock(&mutex);
        item = buffer[out];
        printf("Consumer consumed %d from position %d\n", item, out);
        out = (out + 1) % BUFFER_SIZE;
        pthread_mutex_unlock(&mutex);
        sem_post(&empty);
        consumeItem(item);
        sleep(1);
    }
    pthread_exit(NULL);
}
void initialize()
{
    sem_init(&empty, 0, BUFFER_SIZE);
    sem_init(&full, 0, 0);
    pthread_mutex_init(&mutex, NULL);
}
void destroy()
{
    sem_destroy(&empty);
    sem_destroy(&full);
    pthread_mutex_destroy(&mutex);
}
int main()
{
    pthread_t producerThread;
    pthread_t consumerThread;
    printf("Enter number of items to produce: ");
    scanf("%d", &totalItems);
    initialize();
    pthread_create(&producerThread, NULL, producer, NULL);
    pthread_create(&consumerThread, NULL, consumer, NULL);

```

```
pthread_join(producerThread, NULL);  
pthread_join(consumerThread, NULL);  
destroy();  
printf("\nProducer-Consumer Execution Completed\n");  
return 0;  
}
```

Output:



```
Default: C:\Users\anany\Desktop\OS\prod-cons.exe  
ctrl+alt+1  
C:\Users\anany\Desktop\OS\>Enter number of items to produce: 5  
Producer produced 1 at position 0  
Consumer consumed 1 from position 0  
Consumed Item: 1  
Producer produced 2 at position 1  
Consumer consumed 2 from position 1  
Consumed Item: 2  
Producer produced 3 at position 2  
Consumer consumed 3 from position 2  
Consumed Item: 3  
Producer produced 4 at position 3  
Consumer consumed 4 from position 3  
Consumed Item: 4  
Producer produced 5 at position 4  
Consumer consumed 5 from position 4  
Consumed Item: 5  
  
Producer-Consumer Execution Completed  
  
Process returned 0 (0x0)   execution time : 8.409 s  
Press any key to continue.  
|
```

Observation:

```

Producers Consumers

#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define BUFFER_SIZE 8

int buffer[BUFFER_SIZE];
int in = 0;
int out = 0;
int totalItems;
sem_t empty;
sem_t full;
pthread_mutex_t mutex;

int produceItem(int item) {
    return item;
}

void consumeItem(int item) {
    printf("Consumed Item: %d\n", item);
}

void * producer(void * arg) {
    int item;
    for (int i = 1; i <= totalItems; i++) {
        item = produceItem(i);
        sem_wait(&empty);
        pthread_mutex_lock(&mutex);
        buffer[in] = item;
        printf("Producer produced %d at position %d\n", item, in);
        in = (in + 1) % BUFFER_SIZE;
        pthread_mutex_unlock(&mutex);
        sem_post(&full);
        sleep(1);
    }
    pthread_exit(NULL);
}

void initialize() {
    sem_init(&empty, 0, BUFFER_SIZE);
    sem_init(&full, 0, 0);
    pthread_mutex_init(&mutex, NULL);
}

void destroy() {
    sem_destroy(&empty);
    sem_destroy(&full);
    pthread_mutex_destroy(&mutex);
}

int main() {
    pthread_t producerThread;
    pthread_t consumerThread;
    printf("Enter number of items to produce: ");
    scanf("%d", &totalItems);
    initialize();
    pthread_create(&producerThread, NULL, producer, NULL);
    pthread_create(&consumerThread, NULL, consumeItem, NULL);
    pthread_join(producerThread, NULL);
    pthread_join(consumerThread, NULL);
    destroy();
    printf("Producers-Consumers Execution Completed\n");
    return 0;
}

OUTPUT:
Enter number of items to produce: 32
Producer produced 1 at position 0
Consumer consumed 1 from position 0
Consumed Item: 1
Producer produced 2 at position 1
Consumer consumed 2 from position 1
Consumed Item: 2

```


b) Dining-Philosopher's problem

```
#include <stdio.h>
#include <pthread.h>
#include <unistd.h>
#define MAX 10
int n;
int state[MAX]; // 0 = THINKING, 1 = HUNGRY, 2 = EATING
pthread_mutex_t mutex;
pthread_cond_t cond[MAX];
#define THINKING 0
#define HUNGRY 1
#define EATING 2
int LEFT(int i) { return (i + n - 1) % n; }
int RIGHT(int i) { return (i + 1) % n; }
void test(int i) {
    if (state[i] == HUNGRY &&
        state[LEFT(i)] != EATING &&
        state[RIGHT(i)] != EATING) {
        state[i] = EATING;
        printf("Philosopher %d is EATING\n", i);
        pthread_cond_signal(&cond[i]);
    }
}
void take_fork(int i) {
    pthread_mutex_lock(&mutex);
    state[i] = HUNGRY;
    printf("Philosopher %d is HUNGRY\n", i);
    test(i);
    while (state[i] != EATING)
        pthread_cond_wait(&cond[i], &mutex);
    pthread_mutex_unlock(&mutex);
}
void put_fork(int i) {
    pthread_mutex_lock(&mutex);
    state[i] = THINKING;
    printf("Philosopher %d is THINKING\n", i);
    test(LEFT(i));
    test(RIGHT(i));
    pthread_mutex_unlock(&mutex);
}
```

```

void* philosopher(void* num) {
    int id = *(int*)num;
    while (1) {
        printf("Philosopher %d is THINKING\n", id);
        sleep(1);
        take_fork(id);
        sleep(2);
        put_fork(id);
    }
}

int main() {
    pthread_t p[MAX];
    int ids[MAX];
    printf("Enter number of philosophers: ");
    scanf("%d", &n);
    pthread_mutex_init(&mutex, NULL);
    for (int i = 0; i < n; i++) {
        pthread_cond_init(&cond[i], NULL);
        state[i] = THINKING;
    }
    for (int i = 0; i < n; i++) {
        ids[i] = i;
        pthread_create(&p[i], NULL, philosopher, &ids[i]);
    }
    for (int i = 0; i < n; i++)
        pthread_join(p[i], NULL);
    return 0;
}

```

Output:

```

PS C:\Users\BMSCECSE-L4\Desktop\BM24-41> C:\Users\BMSCECSE-L4\Desktop\BM24-41\din-phil.exe
Enter number of philosophers: 3
Philosopher 1 is THINKING
Philosopher 0 is THINKING
Philosopher 2 is THINKING
Philosopher 0 is HUNGRY
Philosopher 0 is EATING
Philosopher 2 is HUNGRY
Philosopher 1 is HUNGRY
Philosopher 0 is THINKING
Philosopher 2 is EATING
Philosopher 0 is THINKING
Philosopher 0 is HUNGRY
Philosopher 2 is THINKING
Philosopher 1 is EATING
Philosopher 2 is THINKING
Philosopher 2 is HUNGRY
Philosopher 1 is THINKING
Philosopher 0 is EATING
Philosopher 1 is THINKING
Philosopher 1 is HUNGRY

```

Observation:

File	Page	File	Page
Dining Philosophers			
<pre>#include <stdio.h> #include <pthread.h> #include <unistd.h> #define MAX 30 int n; int state[MAX]; pthread_mutex_t mutex; pthread_cond_t cond[MAX]; #define THINKING 0 #define HUNGRY 1 #define EATING 2 int LEFT(int i) { return (i-1) % n; } int RIGHT(int i) { return (i+1) % n; } void test(int i) { if (state[i] == HUNGRY state[LEFT(i)] == EATING state[RIGHT(i)] == EATING) { state[i] = EATING; printf("Philosopher %d is EATING\n", i); pthread_cond_signal(&cond[i]); } } void take_chopstick(int i) { pthread_mutex_lock(&mutex); state[i] = HUNGRY; printf("Philosopher %d is HUNGRY\n", i); test(i); while (state[i] != EATING) {} }</pre>		<pre>pthread_cond_wait(&cond[i], &mutex); pthread_mutex_unlock(&mutex); } void put_chopstick(int i) { pthread_mutex_lock(&mutex); state[i] = THINKING; printf("Philosopher %d is THINKING\n", i); test(LEFT(i)); test(RIGHT(i)); pthread_mutex_unlock(&mutex); } void *philosopher(void *num) { int id = *(int *) num; while (1) { printf("Philosopher %d is THINKING\n", id); sleep(1); take_chopstick(id); sleep(2); put_chopstick(id); } } int main() { pthread_t p[MAX]; int ids[MAX]; printf("Enter the number of philosophers: "); scanf("%d", &n); pthread_mutex_init(&mutex, NULL); for (int i=0; i<n; i++) { pthread_cond_init(&cond[i], NULL); state[i] = THINKING; ids[i] = i; } }</pre>	<pre>pthread_create(&p[i], NULL, philosopher, &ids[i]); } for (int i=0; i<n; i++) {} pthread_join(p[i], NULL); return 0; }</pre>
			OUTPUT: <pre>Enter number of philosophers: 3 Philosopher 1 is THINKING Philosopher 2 is THINKING Philosopher 0 is THINKING Philosopher 2 is HUNGRY Philosopher 2 is EATING Philosopher 0 is HUNGRY Philosopher 0 is HUNGRY Philosopher 2 is THINKING Philosopher 1 is EATING Philosopher 3 is THINKING Philosopher 2 is HUNGRY</pre>

Lab program 5:

Write a C program to simulate:

a) Bankers' algorithm for the purpose of deadlock avoidance. b) Deadlock Detection

a) Bankers' algorithm for the purpose of deadlock avoidance.

```
#include <stdio.h>
int main()
{
    int n, m, i, j, k;
    printf("Enter number of processes and resources: ");
    scanf("%d %d", &n, &m);
    int alloc[n][m], max[n][m], need[n][m];
    int avail[m], finish[n], safe[n];
    printf("Enter Allocation Matrix:\n");
    for(i = 0; i < n; i++)
        for(j = 0; j < m; j++)
            scanf("%d", &alloc[i][j]);
    printf("Enter Maximum Matrix:\n");
    for(i = 0; i < n; i++)
        for(j = 0; j < m; j++)
            scanf("%d", &max[i][j]);
    printf("Enter Available Resources:\n");
    for(i = 0; i < m; i++)
        scanf("%d", &avail[i]);
    for(i = 0; i < n; i++)
        for(j = 0; j < m; j++)
            need[i][j] = max[i][j] - alloc[i][j];
    for(i = 0; i < n; i++)
        finish[i] = 0;
    int count = 0;
    while(count < n)
    {
        int found = 0;
        for(i = 0; i < n; i++)
        {
            if(finish[i] == 0)
            {
                for(j = 0; j < m; j++)
                    if(need[i][j] > avail[j])
                        break;
            }
        }
        if(found == 0)
            continue;
        for(j = 0; j < m; j++)
            avail[j] -= need[i][j];
        finish[i] = 1;
        count++;
    }
}
```

```

        if(j == m)
        {
            for(k = 0; k < m; k++)
                avail[k] += alloc[i][k];
            safe[count++] = i;
            finish[i] = 1;
            found = 1;
        }
    }
}
if(found == 0)
{
    printf("System is NOT in safe state\n");
    return 0;
}
printf("System is in SAFE state\nSafe Sequence: ");
for(i = 0; i < n; i++)
    printf("P%d ", safe[i]);
return 0;
}

```

Output:

```

PS C:\Users\BMSCECSE-L4\Desktop\BM24-41> C:\Users\BMSCECSE-L4\Desktop\BM24-41\bankers.exe
Enter number of processes and resources: 3 2
Enter Allocation Matrix:
0 1
2 0
1 1
Enter Maximum Matrix:
1 2
2 1
3 2
Enter Available Resources:
1 1
System is in SAFE state
Safe Sequence: P0 P1 P2
PS C:\Users\BMSCECSE-L4\Desktop\BM24-41> █

```

Observation:

```

Banker's Algorithm (Deadlock)

#include <stdio.h>
int main() {
    int n, m, i, j, k;
    printf("Enter number of processes and resources\n");
    scanf("%d %d", &n, &m);
    int alloc[n][m], max[n][m], need[n][m];
    int avail[m], finish[n], safe[n];
    printf("Enter Allocation Matrix :\n");
    for (i=0; i<n; i++)
        for (j=0; j<m; j++)
            scanf("%d", &alloc[i][j]);
    printf("Enter Maximum Matrix :\n");
    for (i=0; i<n; i++)
        for (j=0; j<m; j++)
            scanf("%d", &max[i][j]);
    printf("Enter Available Resource :\n");
    for (i=0; i<m; i++)
        scanf("%d", &avail[i]);
    for (i=0; i<n; i++)
        finish[i]=0;
    int count = 0;
    while (count < n) {
        int found = 0;
        for (i=0; i<n; i++) {
            if (finish[i]==0) {
                for (j=0; j<m; j++)
                    if (need[i][j] > avail[j])
                        break;
                if (j==m) {
                    for (k=0; k<m; k++)
                        avail[k] += alloc[i][k];
                }
            }
        }
        if (found==0) {
            printf("System is NOT in safe state\n");
            return 0;
        }
        printf("System is in SAFE state\n Safe Sequence:");
        for (i=0; i<n; i++)
            printf("P%d", safe[i]);
        return 0;
    }
}

OUTPUT:
Enter number of processes and resources: 3 2
Enter Allocation Matrix: 0 1
                        8 0
                        1 1
Enter Maximum Matrix: 1 2
                      2 1
                      3 2
Enter Available Resource: 1 1
System is in safe state.
Safe Sequence: P0 P1 P2

```

b) Deadlock Detection

```
#include <stdio.h>

int main() {
    int n, m;
    // n = number of processes
    // m = number of resource types
    printf("Enter number of processes: ");
    scanf("%d", &n);
    printf("Enter number of resource types: ");
    scanf("%d", &m);
    int alloc[n][m], request[n][m], avail[m];
    int finish[n];
    printf("\nEnter Allocation Matrix:\n");
    for(int i = 0; i < n; i++) {
        for(int j = 0; j < m; j++) {
            scanf("%d", &alloc[i][j]);
        }
    }
    printf("\nEnter Request Matrix:\n");
    for(int i = 0; i < n; i++) {
        for(int j = 0; j < m; j++) {
            scanf("%d", &request[i][j]);
        }
    }
    printf("\nEnter Available Resources:\n");
    for(int i = 0; i < m; i++) {
        scanf("%d", &avail[i]);
    }
    for(int i = 0; i < n; i++) {
        finish[i] = 0;
    }
    int work[m];
    for(int i = 0; i < m; i++) {
        work[i] = avail[i];
    }
    int found;
    do {
        found = 0;
        for(int i = 0; i < n; i++) {
```



```

        if(finish[i] == 0) {
            int canExecute = 1;
            for(int j = 0; j < m; j++) {
                if(request[i][j] > work[j]) {
                    canExecute = 0;
                    break;
                }
            }
            if(canExecute) {
                for(int j = 0; j < m; j++) {
                    work[j] += alloc[i][j];
                }
                finish[i] = 1;
                found = 1;
                printf("\nProcess P%d executed successfully.", i);
            }
        }
    }
} while(found);
int deadlock = 0;
printf("\n\nDeadlocked Processes: ");
for(int i = 0; i < n; i++) {
    if(finish[i] == 0) {
        printf("P%d ", i);
        deadlock = 1;
    }
}
if(deadlock == 0) {
    printf("None");
    printf("\nSystem is NOT in deadlock.");
} else {
    printf("\nSystem IS in deadlock.");
}
return 0;
}

```

Output:

```
PS C:\Users\BMSCECSE-L4\Desktop\BM24-41> C:\Users\BMSCECSE-L4\Desktop\BM24-41\deadlock-dection.exe
Enter number of processes: 5
Enter number of resource types: 3

Enter Allocation Matrix:
0 1 0
2 0 0
3 0 3
2 1 1
0 0 2

Enter Request Matrix:
0 0 0
2 0 2
0 0 0
1 0 0
0 0 2

Enter Available Resources:
0 0 0

Process P0 executed successfully.
Process P2 executed successfully.
Process P3 executed successfully.
Process P4 executed successfully.
Process P1 executed successfully.

Deadlocked Processes: None
System is NOT in deadlock.
PS C:\Users\BMSCECSE-L4\Desktop\BM24-41> |
```

Observation:

Deadlock Detection

```

#include <stdio.h>
int main() {
    int m, n;
    // n -> no. of processes
    // m -> no. of resource types
    printf("Enter number of processes: ");
    scanf("%d", &n);
    printf("Enter number of resource types: ");
    scanf("%d", &m);
    int alloc[n][m], request[n][m], avail[m];
    int finish[n];
    printf("\n Enter Allocation Matrix: \n");
    for (int i=0; i<n; i++) {
        for (int j=0; j<m; j++) {
            scanf("%d", &alloc[i][j]);
        }
    }
    printf("\n Enter Request Matrix: \n");
    for (int i=0; i<n; i++) {
        for (int j=0; j<m; j++) {
            scanf("%d", &request[i][j]);
        }
    }
    printf("\n Enter Available Resource: \n");
    for (int i=0; i<m; i++) {
        scanf("%d", &avail[i]);
    }
    for (int i=0; i<n; i++) {
        finish[i] = 0;
    }

    if (deadlock == 0) {
        printf("None");
        printf("\n System is NOT in deadlock.");
    } else {
        printf("\n System IS in deadlock.");
    }
    return 0;
}

```

OUTPUT:

Enter number of processes: 5
 Enter number of resource types: 3
 Enter Allocation Matrix:
 0 0 0
 2 0 0
 3 0 3
 2 1 1
 0 0 2

Enter Request Matrix:
 0 0 0
 2 0 2
 0 0 0
 1 0 0
 0 0 2

Enter Available Resources:
 0 0 0

Process P0 executed successfully
 Process P2 executed successfully
~~Deadlock Processes: P1, P3, P4~~
 System IS in deadlock
 Process P3 executed successfully
 Process P4 executed successfully

```

for (int i=0; i<m; i++) {
    work[i] = avail[i];
}
int found;
do {
    found = 0;
    for (int i=0; i<n; i++) {
        if (finish[i] == 0) {
            int countExecute = 0;
            for (int j=0; j<m; j++) {
                if (request[i][j] > work[j]) {
                    countExecute = 0;
                    break;
                }
            }
            if (countExecute) {
                for (int j=0; j<m; j++) {
                    work[j] += alloc[i][j];
                }
                finish[i] = 1;
                found = 1;
                printf("\n Process %d executed successfully.", i);
            }
        }
    }
    while (!found);
    int deadlock = 0;
    printf("\n Deadlock Processes: ");
    for (int i=0; i<n; i++) {
        if (finish[i] == 0) {
            // ...
        }
    }
}

```

Process P1 executed successfully
 Deadlock Processes: None
 System is NOT in deadlock.

27/5/24

Lab program 6:

Write a C program to simulate the following contiguous memory allocation techniques.

a) Worst-fit b) Best-fit c) First-fit

a) Worst-fit

```
#include <stdio.h>

int main() {
    int nb, np, i, j;
    printf("Enter number of memory blocks: ");
    scanf("%d", &nb);
    int block[nb];
    printf("Enter sizes of memory blocks:\n");
    for(i = 0; i < nb; i++) {
        scanf("%d", &block[i]);
    }
    printf("Enter number of processes: ");
    scanf("%d", &np);
    int process[np], allocation[np];
    printf("Enter sizes of processes:\n");
    for(i = 0; i < np; i++) {
        scanf("%d", &process[i]);
        allocation[i] = -1;
    }
    for(i = 0; i < np; i++) {
        int worstIdx = -1;
        for(j = 0; j < nb; j++) {
            if(block[j] >= process[i]) {
                if(worstIdx == -1 ||
                    block[j] > block[worstIdx]) {
                    worstIdx = j;
                }
            }
        }
        if(worstIdx != -1) {
            allocation[i] = worstIdx;
            block[worstIdx] -= process[i];
        }
    }
    printf("\nWorst Fit Memory Allocation\n");
```

```

printf("Process\tProcess Size\tBlock Allocated\n");
for(i = 0; i < np; i++) {
    printf("P%d\t%d\t", i + 1, process[i]);
    if(allocation[i] != -1)
        printf("%d\n", allocation[i] + 1);
    else
        printf("Not Allocated\n");
}
return 0;
}

```

Output:

```

C:\Users\anany\Desktop\OS\worstfit.exe
Enter number of memory blocks: 5
Enter sizes of memory blocks:
100
500
200
300
600
Enter number of processes: 4
Enter sizes of processes:
212
417
112
426

Worst Fit Memory Allocation
Process Process Size    Block Allocated
P1      212              5
P2      417              2
P3      112              5
P4      426              Not Allocated

Process returned 0 (0x0)   execution time : 27.468 s
Press any key to continue.

```

b) Best-fit

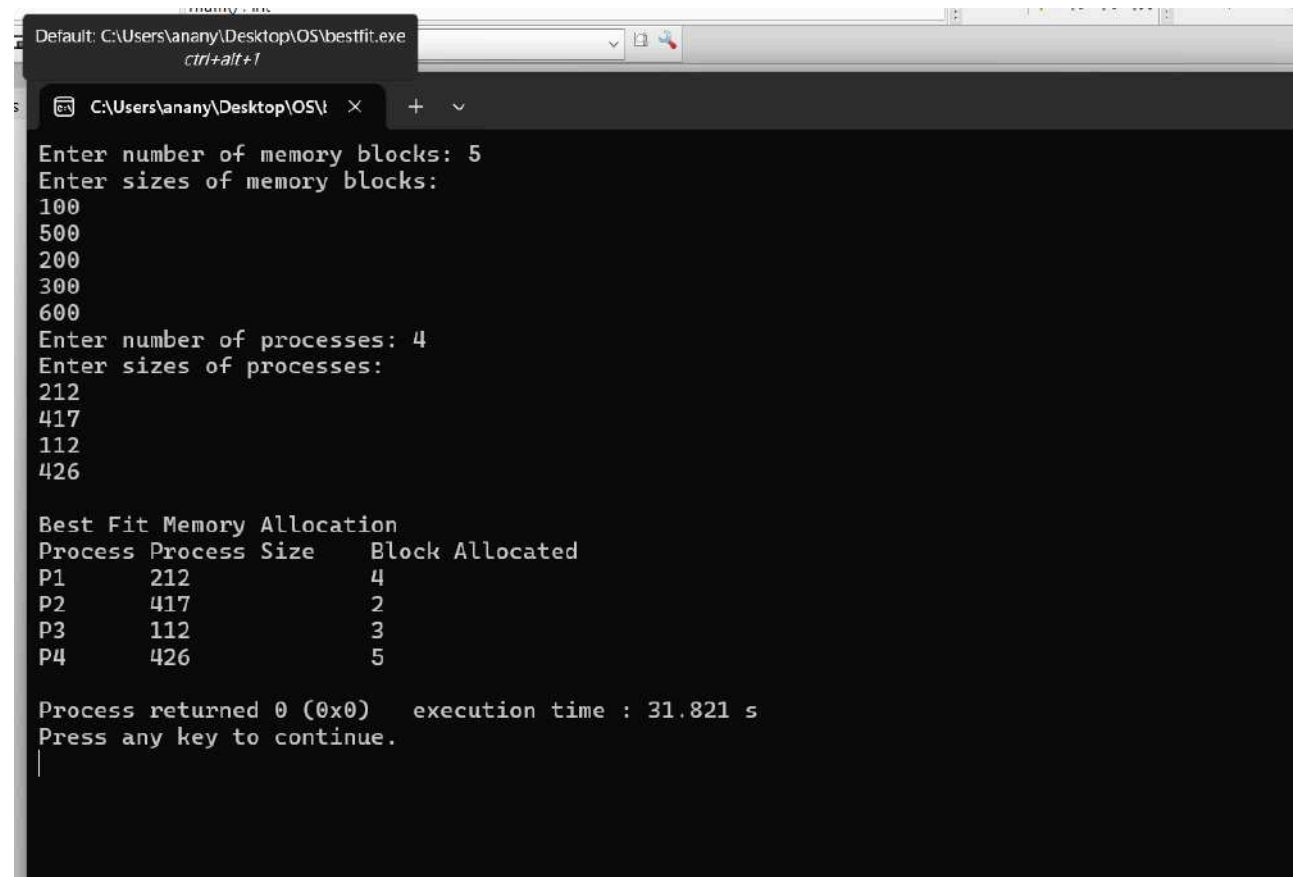
```
#include <stdio.h>
int main() {
    int nb, np, i, j;
    printf("Enter number of memory blocks: ");
    scanf("%d", &nb);
    int block[nb];
    printf("Enter sizes of memory blocks:\n");
    for(i = 0; i < nb; i++) {
        scanf("%d", &block[i]);
    }
    printf("Enter number of processes: ");
    scanf("%d", &np);
    int process[np], allocation[np];
    printf("Enter sizes of processes:\n");
    for(i = 0; i < np; i++) {
        scanf("%d", &process[i]);
        allocation[i] = -1;
    }
    for(i = 0; i < np; i++) {
        int bestIdx = -1;
        for(j = 0; j < nb; j++) {
            if(block[j] >= process[i]) {
                if(bestIdx == -1 ||
                    block[j] < block[bestIdx]) {
                    bestIdx = j;
                }
            }
        }
        if(bestIdx != -1) {
            allocation[i] = bestIdx;
            block[bestIdx] -= process[i];
        }
    }
    printf("\nBest Fit Memory Allocation\n");
    printf("Process\tProcess Size\tBlock Allocated\n");
    for(i = 0; i < np; i++) {
        printf("P%d\t%d\t\t", i + 1, process[i]);
        if(allocation[i] != -1)
            printf("%d\n", allocation[i] + 1);
    }
}
```

```

        else
            printf("Not Allocated\n");
    }
    return 0;
}

```

Output:



```

Default: C:\Users\anany\Desktop\OS\bestfit.exe
ctrl+alt+l

C:\Users\anany\Desktop\OS\i x + v

Enter number of memory blocks: 5
Enter sizes of memory blocks:
100
500
200
300
600
Enter number of processes: 4
Enter sizes of processes:
212
417
112
426

Best Fit Memory Allocation
Process Process Size    Block Allocated
P1      212           4
P2      417           2
P3      112           3
P4      426           5

Process returned 0 (0x0)   execution time : 31.821 s
Press any key to continue.
|

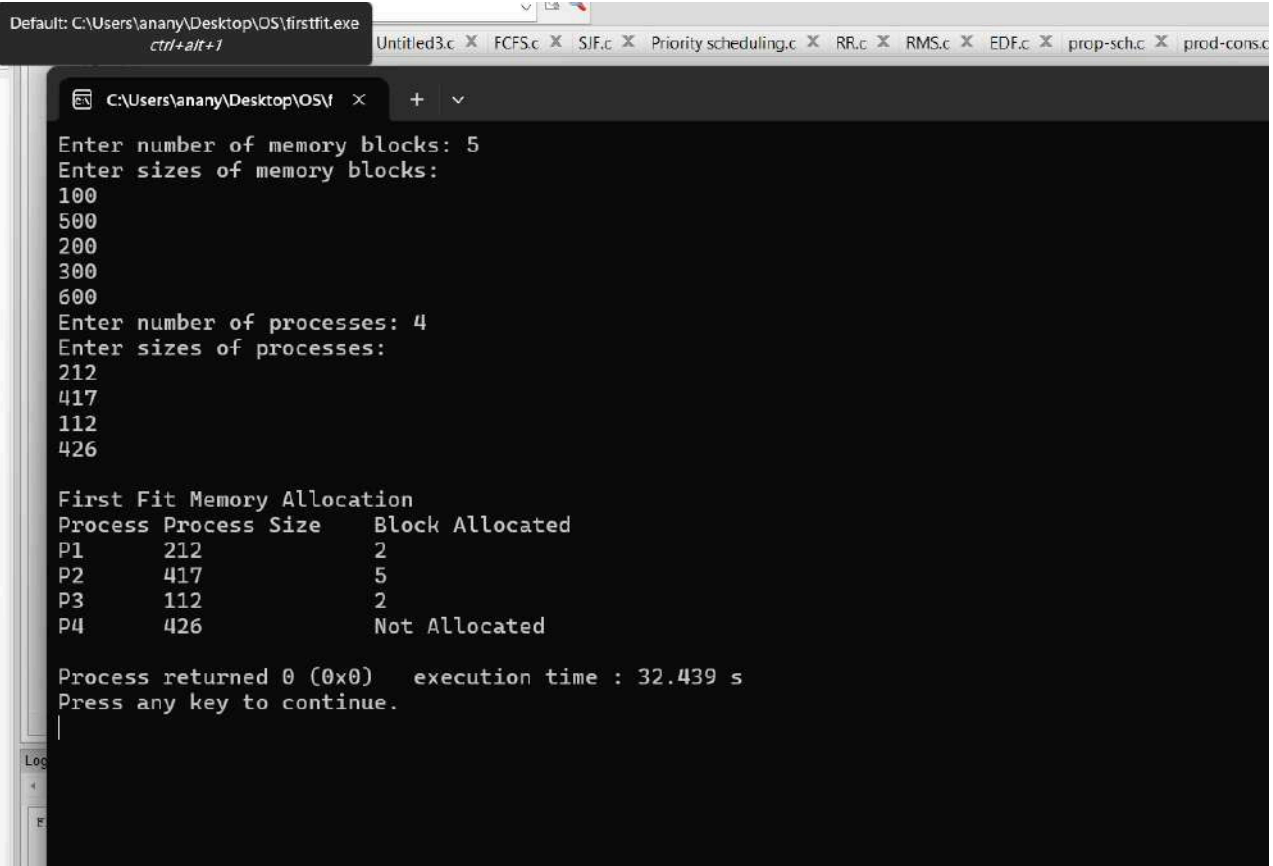
```


c) First-fit

```
#include <stdio.h>

int main() {
    int nb, np, i, j;
    printf("Enter number of memory blocks: ");
    scanf("%d", &nb);
    int block[nb];
    printf("Enter sizes of memory blocks:\n");
    for(i = 0; i < nb; i++) {
        scanf("%d", &block[i]);
    }
    printf("Enter number of processes: ");
    scanf("%d", &np);
    int process[np], allocation[np];
    printf("Enter sizes of processes:\n");
    for(i = 0; i < np; i++) {
        scanf("%d", &process[i]);
        allocation[i] = -1;
    }
    for(i = 0; i < np; i++) {
        for(j = 0; j < nb; j++) {
            if(block[j] >= process[i]) {
                allocation[i] = j;
                block[j] -= process[i];
                break;
            }
        }
    }
    printf("\nFirst Fit Memory Allocation\n");
    printf("Process\tProcess Size\tBlock Allocated\n");
    for(i = 0; i < np; i++) {
        printf("P%d\t%d\t\t", i + 1, process[i]);
        if(allocation[i] != -1)
            printf("%d\n", allocation[i] + 1);
        else
            printf("Not Allocated\n");
    }
    return 0;
}
```

Output:



```
Default: C:\Users\anany\Desktop\OS\firstfit.exe
ctrl+alt+1

Enter number of memory blocks: 5
Enter sizes of memory blocks:
100
500
200
300
600
Enter number of processes: 4
Enter sizes of processes:
212
417
112
426

First Fit Memory Allocation
Process Process Size    Block Allocated
P1      212             2
P2      417             5
P3      112             2
P4      426            Not Allocated

Process returned 0 (0x0)    execution time : 32.439 s
Press any key to continue.
```

Lab program 7:

Write a C program to simulate page replacement algorithms.

a) FIFO b) LRU c) Optimal

a) FIFO

```
#include <stdio.h>
int main() {
    int pages, frames, i, j;
    printf("Enter number of pages: ");
    scanf("%d", &pages);
    int page[pages];
    printf("Enter page reference string:\n");
    for(i = 0; i < pages; i++) {
        scanf("%d", &page[i]);
    }
    printf("Enter number of frames: ");
    scanf("%d", &frames);
    int frame[frames];
    for(i = 0; i < frames; i++)
        frame[i] = -1;
    int pageFaults = 0;
    int index = 0;
    printf("\nPage\tFrames\n");
    for(i = 0; i < pages; i++) {
        int found = 0;
        for(j = 0; j < frames; j++) {
            if(frame[j] == page[i]) {
                found = 1;
                break;
            }
        }
        if(!found) {
            frame[index] = page[i];
            index = (index + 1) % frames;
            pageFaults++;
        }
        printf("%d\t", page[i]);
        for(j = 0; j < frames; j++) {
            if(frame[j] != -1)
                printf("%d ", frame[j]);
            else
```

```

        printf("- ");
    }
    printf("\n");
}
printf("Total Page Faults = %d\n", pageFaults);
return 0;
}

```

Output:

```

Default: C:\Users\anany\Desktop\OS\lifo.exe
ctrl+alt+1

C:\Users\anany\Desktop\OS\lifo.exe
Enter number of pages: 12
Enter page reference string:
1 2 3 4 1 2 5 1 2 3 4 5
Enter number of frames: 3

Page    Frames
1       1 - -
2       1 2 -
3       1 2 3
4       4 2 3
1       4 1 3
2       4 1 2
5       5 1 2
1       5 1 2
2       5 1 2
3       5 3 2
4       5 3 4
5       5 3 4
Total Page Faults = 9

Process returned 0 (0x0)   execution time : 29.079 s
Press any key to continue.

```

b) LRU

```
#include <stdio.h>
int main() {
    int n, frames, i, j, k;
    printf("Enter number of pages: ");
    scanf("%d", &n);
    int pages[n];
    printf("Enter page reference string:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }
    printf("Enter number of frames: ");
    scanf("%d", &frames);
    int frame[frames], recent[frames];
    for(i = 0; i < frames; i++) {
        frame[i] = -1;
        recent[i] = 0;
    }
    int pageFaults = 0, time = 0;
    printf("\nPage\tFrames\n");
    for(i = 0; i < n; i++) {
        int found = 0;
        for(j = 0; j < frames; j++) {
            if(frame[j] == pages[i]) {
                found = 1;
                recent[j] = ++time;
                break;
            }
        }
        if(!found) {
            int lruIndex = 0;
            for(j = 0; j < frames; j++) {
                if(frame[j] == -1) {
                    lruIndex = j;
                    break;
                }
            }
            if(recent[j] < recent[lruIndex])
                lruIndex = j;
        }
    }
}
```

```

        frame[lruIndex] = pages[i];
        recent[lruIndex] = ++time;
        pageFaults++;
    }
    printf("%d\t", pages[i]);
    for(k = 0; k < frames; k++) {
        if(frame[k] != -1)
            printf("%d ", frame[k]);
        else
            printf("- ");
    }
    printf("\n");
}
printf("Total Page Faults = %d\n", pageFaults);
return 0;
}

```

Output:

```

C:\Users\anany\Desktop\OS\lru.exe
ctrl+alt+1

Enter number of pages: 12
Enter page reference string:
1 2 3 4 1 2 5 1 2 3 4 5
Enter number of frames: 3

Page    Frames
1       1 - -
2       1 2 -
3       1 2 3
4       4 2 3
1       4 1 3
2       4 1 2
5       5 1 2
1       5 1 2
2       5 1 2
3       3 1 2
4       3 4 2
5       3 4 5
Total Page Faults = 10

Process returned 0 (0x0)   execution time : 18.594 s
Press any key to continue.
|

```

c) Optimal

```
#include <stdio.h>
int main() {
    int n, frames, i, j, k;
    printf("Enter number of pages: ");
    scanf("%d", &n);
    int pages[n];
    printf("Enter page reference string:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }
    printf("Enter number of frames: ");
    scanf("%d", &frames);
    int frame[frames];
    for(i = 0; i < frames; i++)
        frame[i] = -1;
    int pageFaults = 0;
    printf("\nPage\tFrames\n");
    for(i = 0; i < n; i++) {
        int found = 0;
        for(j = 0; j < frames; j++) {
            if(frame[j] == pages[i]) {
                found = 1;
                break;
            }
        }
        if(!found) {
            int replaceIndex = -1;
            for(j = 0; j < frames; j++) {
                if(frame[j] == -1) {
                    replaceIndex = j;
                    break;
                }
            }
            if(replaceIndex == -1) {
                int farthest = -1;
                for(j = 0; j < frames; j++) {
                    int nextUse;
                    for(nextUse = i + 1; nextUse < n; nextUse++) {
                        if(frame[j] == pages[nextUse])
```



```

        break;
    }
    if(nextUse == n) {
        replaceIndex = j;
        break;
    }
    if(nextUse > farthest) {
        farthest = nextUse;
        replaceIndex = j;
    }
}
frame[replaceIndex] = pages[i];
pageFaults++;
}
printf("%d\t", pages[i]);
for(k = 0; k < frames; k++) {
    if(frame[k] != -1)
        printf("%d ", frame[k]);
    else
        printf("- ");
}
printf("\n");
}
printf("Total Page Faults = %d\n", pageFaults);
return 0;
}

```

Output:

```

Default: C:\Users\anany\Desktop\OS\optimal.exe
ctrl+alt+t

C:\Users\anany\Desktop\OS> Enter number of pages: 12
Enter page reference string:
1 2 3 4 1 2 5 1 2 3 4 5
Enter number of frames: 3

Page    Frames
1       1 - -
2       1 2 -
3       1 2 3
4       1 2 4
1       1 2 4
2       1 2 4
5       1 2 5
1       1 2 5
2       1 2 5
3       3 2 5
4       4 2 5
5       4 2 5
Total Page Faults = 7

```